

Comprehensive Skills, Memory & Data Reference

Generated: 2026-08-11 | Consolidated system reference for skills, memory policy, and related data

1. Available Skills Overview

Skill	Primary Use
color	Color palettes, accessibility/contrast, shade scales, UI color auditing
docx	Create/read/edit Word documents (.docx / .dotx)
ffmpeg	Media processing: convert, trim, resize, compress, GIFs, subtitles, combine videos
finance	Financial data and calculations
image-gen-edit	Image generation and editing workflows
imagemagick	ImageMagick-based image processing
mcp	MCP-related tooling
memory-edit	Policy for storing, updating, deleting user personal memory
pdf	Create, read, transform, OCR, form-fill PDFs
pptx	PowerPoint / presentation creation and editing
skill-creator	Create and update custom skills
skill-installer	Install skills from GitHub into .grok/skills/
tasks	Scheduled tasks, reminders, recurring Grok actions
xlsx	Spreadsheet creation, editing, cleaning (.xlsx, .csv, etc.)

2. Memory Retention & Edit Policy

Source: *memory-edit skill*

Store — Durable personal facts worth remembering

- Identity & demographics (name, age, birthday, pronouns, nationality, languages)
- Relationships & family (partner, children, pets)
- Location (current city, neighborhood)
- Health & constraints (allergies, accessibility needs, medication)
- Work & education (occupation, career history, degree)
- Plans & goals (major life events, long-term projects)
- Preferences (dietary, communication style, units, favorite genres)
- Hobbies & interests; Religion & spirituality; Cultural identity
- Political affiliation; Vehicles; Living situation; Physical attributes
- Personal milestones & life history; General financial context

Do NOT Store

- Ephemeral states ("I'm tired today")
- World knowledge / factual questions
- Opinions about external topics (unless strong preference)

- Information about third parties (unless directly relevant)
- Hypotheticals, jokes, sarcasm
- Illegal, harmful, or false content
- Sensitive credentials (passwords, API keys, SSN, bank details, etc.)
- Session-scoped tasks; Roleplay / fictional persona details
- Delegated queries ("My friend wants to know...")

Entry Format & Update Rules

One atomic fact per entry with short factual phrasing and a date/timestamp (e.g. - Lives in Austin [2025-03-25]). Never merge unrelated facts. Check for duplicates before adding. **Add** new durable facts or explicit remember requests; **Replace** when facts change or are corrected; **Delete** immediately on explicit user request (hard delete).

3. Skill Creation Guidelines

Source: *skill-creator skill*

Skills are modular instruction packages that specialize the agent for specific tasks, domains, or workflows. Do not create skills for knowledge the model already possesses.

Directory Structure

```
skill-name/
■■■■ SKILL.md           # Required: frontmatter + instructions
■■■■ scripts/          # Optional: executable code
■■■■ references/        # Optional: docs loaded on demand
■■■■ assets/           # Optional: templates, images, fonts
```

Frontmatter Rules

- **name:** kebab-case, must match directory name, 2–64 characters
- **description:** primary trigger mechanism (max 1024 chars). Include what and when. Avoid colon-space, <, >. Single-line plain YAML scalar.
- **Body:** concise imperative instructions. Target < 500 lines; move excess to references/.

Progressive Disclosure

1. Metadata (name + description) — always visible (~100 tokens)
2. SKILL.md body — loaded on demand (target < 5k tokens)
3. Bundled resources — read by the model as needed (unlimited)

4. Detailed Skill Summaries

color — Color palettes, accessibility/contrast checking, shade scales, perceptual safety, UI color auditing.

docx — Create, read, edit Word documents (.docx/.dotx). Reports, memos, letters, templates, tracked changes, comments. Not for PDFs or spreadsheets.

ffmpeg — Inspect, convert, trim, resize, compress, extract frames/audio, mute, GIFs, subtitles, overlays, combine videos via ffmpeg/ffprobe.

memory-edit — Decide what to store/update/delete in user memory.md. Consult on personal facts or explicit remember/forget requests.

pdf — Text/table extraction, merge/split, rotate, watermark, encrypt, OCR scanned docs, fill forms. Uses pypdf, pdfplumber, reportlab, poppler, qpdf.

pptx — Create, read, edit, combine, or split PowerPoint presentations (.pptx).

skill-creator — Create and manage modular skills that extend agent capabilities with domain procedures and tool integrations.

skill-installer — Install skills from GitHub repositories into .grok/skills/, persisted across sessions.

tasks — Create, modify, view, delete, pause, resume scheduled tasks and reminders; retrieve task results.
xlsx — Spreadsheet work where .xlsx/.csv/.tsv is primary input or output: formulas, charts, cleaning, formatting.
Other — finance, image-gen-edit, imagemagick, mcp — domain-specific tooling; consult their individual SKILL.md files.

5. Environment Notes

- Working directory: /home/workdir/artifacts
- Skills location: /root/.grok/skills/
- Internet access enabled in the sandbox
- PDF generation uses reportlab (Platypus for flowing documents, Canvas for absolute layout)
- Prefer existing skills over reinventing procedures

6. How to Use This Reference

1. Identify the relevant skill from the overview table.
2. Read the corresponding /root/.grok/skills/<name>/SKILL.md for full instructions.
3. For personal facts, apply the Memory Retention Policy before any write.
4. When creating new capabilities, follow the skill-creator structure exactly.